

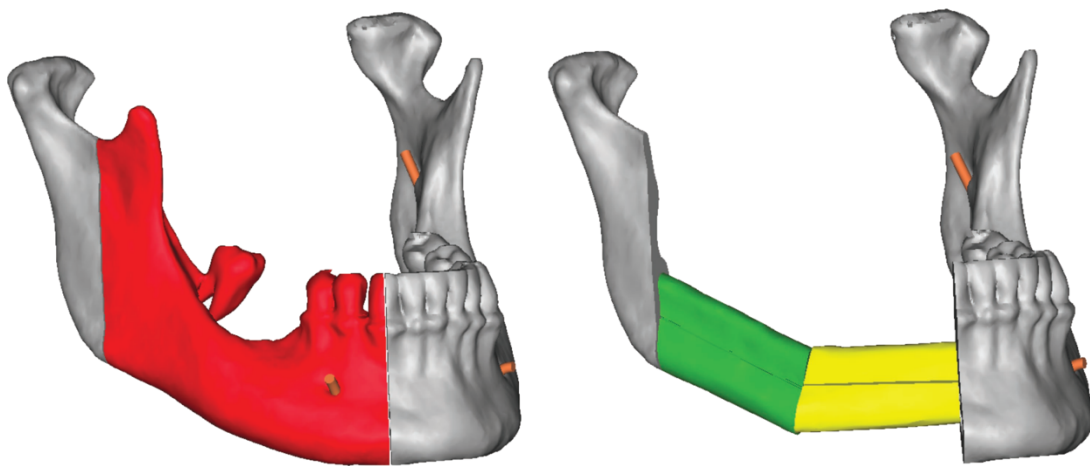
Mandibular Bone Union Detection

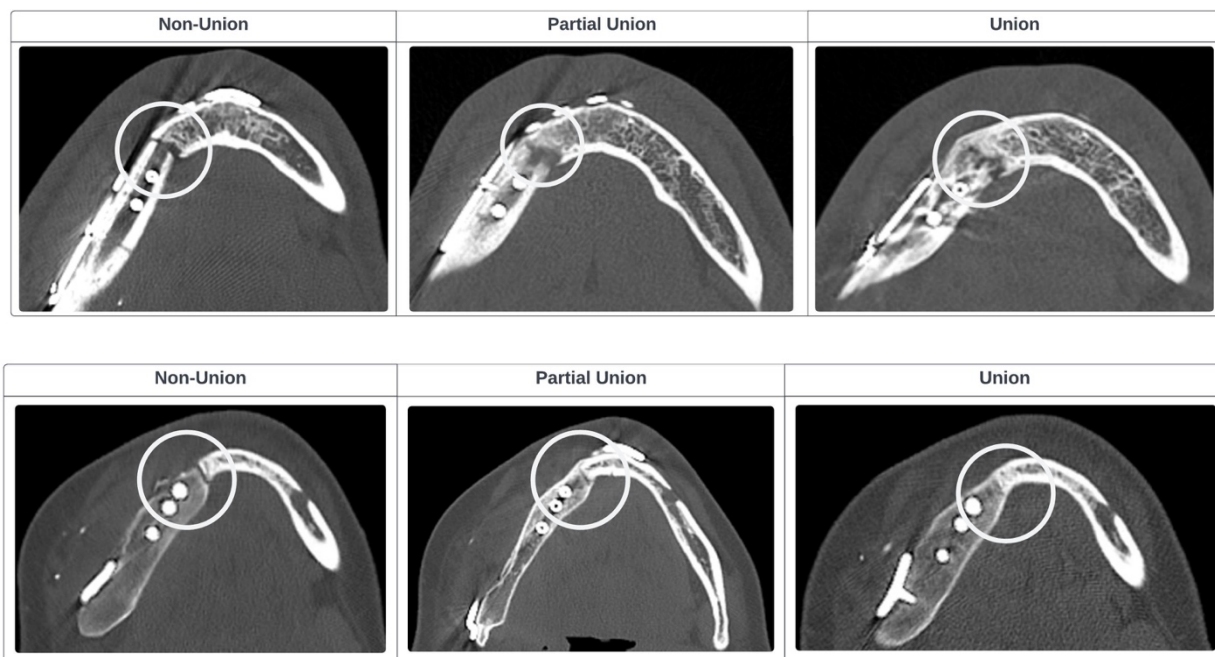
Technical Project Description

1. Background

The mandible is the lower jaw — the bone that holds the lower teeth and moves when we chew and speak. Sometimes a piece of it has to be removed: it may be badly broken in a car accident, or a tumour may have grown into it and the surgeon has to cut away the affected part. That leaves a gap in the jaw, and the gap has to be filled so the patient can eat and speak again. The usual solution is to take a piece of bone from somewhere else in the same patient, most often the fibula (one of the two bones in the lower leg), cut it to the right shape, and screw it into the gap with small metal plates. This transplanted piece of bone is called the **graft**, and the places where it touches the patient's original jaw bone are called the **osteotomy sites**. They are simply the joins between the new bone and the old bone.

After the surgery, the two bones are only held together by the metal plates. Over the following months the body is supposed to grow new bone across each join, so that the graft and the original jaw fuse into one solid piece. This is called bone union, and it can be thought of as how much of the join has actually filled in with new bone, expressed as a percentage. Sometimes it does not happen, and the jaw stays permanently unstable, which is painful and usually needs another operation. Doctors therefore check the joins regularly using medical images. A normal X-ray flattens everything into one picture, so the metal plates and other bones tend to sit on top of the join and hide it. A CT scan is better: it produces a stack of cross-sectional images through the patient, so each join can be looked at one image at a time. Today a doctor does this by hand — scrolling through hundreds of images to find the few that show a join, then judging each one by eye. It is slow, it has to be repeated at every follow-up visit, and two doctors looking at the same scan often disagree. The goal of this project is to automate the first step of that process: finding where the joins are in the scan.





2. Project objective

The objective of this project is to design and implement a pipeline that automatically detects the osteotomy sites of a reconstructed mandible in post-operative CT images. Locating these regions of interest is the prerequisite step for any subsequent quantitative assessment of bone union, and it is the step this project targets.

The work is divided into two parts, followed by a short written discussion and a brief survey.

3. Part 1 — Understanding the CT data

The purpose of this part is to become familiar with the structure and the conventions of CT volumes before any modelling is attempted.

3.1 Anatomical viewing planes

Write a short explanation of the three standard anatomical planes used to view a CT volume: the axial, the coronal and the sagittal plane.

3.2 Analysis of the provided CT volume

A single CT volume is provided. Load it programmatically and report the following, together with the code used to produce each answer:

- The shape of the loaded image array, with an explanation of what each dimension represents.
- The minimum and the maximum intensity value present in the volume, and a brief comment on what these values mean in the context of CT intensity units.
- The number of slices the volume contains. The provided volume is stored in the axial orientation, so this is the number of axial slices.
- A small number of representative slices exported as JPG images and included with the submission.

4. Part 2 — Osteotomy site detection pipeline

In this part an object detection pipeline is implemented and evaluated on a dataset of annotated CT slices.

4.1 Provided dataset

To keep the task tractable, the relevant slices have already been selected from the original volumes and exported as JPG images. The dataset contains two directories:

Directory	Contents
images	Axial CT slices exported as JPG images.
labels	One annotation file per image, in plain text, containing the bounding box annotations of the osteotomy sites in that slice.

Every image has exactly one corresponding label file with the same base name. File names follow the convention:

patientID_sessionNumber_sliceNumber.jpg / .txt

Field	Meaning
patientID	Identifier of the patient the scan belongs to.
sessionNumber	Index of the imaging session. A patient may be scanned at several points in time during follow-up; each such acquisition is one session.
sliceNumber	Index of the axial slice extracted from the original volume of that session.

For example, 920_1_43.jpg is axial slice 43 taken from the first imaging session of patient 920, and 920_1_43.txt holds its annotations.

4.2 Accessing the dataset

The dataset is hosted on Google Drive at the link below. Access is restricted, so please open the link and submit an access request for it; permission will be granted to you individually.

[link](#)

The data is provided for the purposes of this project only: it should not be redistributed, uploaded elsewhere, or included in any public repository or published material.

4.3 Tasks

- Inspect the dataset and summarise it: number of images, number of patients and sessions.
- Define the training, validation and test splits, and justify how they were constructed.
- Train a detection model on the provided data. Any framework or architecture may be used.
- Report the evaluation, and analyse the results.
- Include qualitative results: example slices with predicted and ground-truth boxes overlaid, covering both successful and failed cases.

4.4 What is expected in the write-up

The reasoning behind each choice is more important than the final numbers. The results should then be analysed rather than merely reported: describe where the model performs well, where it fails, and what the failure modes suggest about the data or the approach.

The report should not be long, but it should be comprehensive. Clarity and sound reasoning are valued above the absolute performance of the model. A follow-up technical discussion will be arranged, in which the submitted solution and the decisions behind it will be presented and discussed.

5. Discussion questions

Answer the following two questions in the report.

- **Limitations.** What are the limitations of this project?
- **Improvements.** How could the results of this project be improved?

6. Short survey

Please answer the following three questions. They are used only to understand your interests and do not affect the assessment of your work. Rate each item on a scale from 0 (not interested at all) to 5 (extremely interested).

- How interested are you in computer vision? (0 – 5): _____
- How interested are you in medical image processing? (0 – 5): _____
- How interested were you in this project? (0 – 5): _____